

Achieving transportation equity: Meaningful public involvement to meet the needs of underserved communities

Alex Karner (corresponding author)

Graduate Program in Community and Regional Planning
School of Architecture
The University of Texas at Austin
310 Inner Campus Drive B7500
Austin, TX 78712-1009
alex.karner@utexas.edu
+1-512-471-5663

Richard A. Marcantonio

Public Advocates Inc.
131 Steuart Street, Suite 300
San Francisco, CA 94105
Rmarcantonio@publicadvocates.org
+1-415-431-7430 ext. 308

Paper published in *Public Works Management & Policy*.

Note: This is the author's version of a work that was accepted for publication in *Public Works Management & Policy*. Changes resulting from the publishing process, such as peer review, editing, corrections, structural formatting, and other quality control mechanisms may not be reflected in this document. Changes may have been made to this work since it was submitted for publication.

Print version available at: <http://journals.sagepub.com/doi/full/10.1177/1087724X17738792>.

© 2017. This manuscript version is made available under the CC-BY-NC-ND 4.0 license 24
<http://creativecommons.org/licenses/by-nc-nd/4.0/> 25

Abstract

Civil rights and environmental justice laws and regulations aim to ensure fair processes and outcomes in regional transportation planning. Despite these requirements, people of color and low-income tend to be disproportionately burdened while receiving few benefits. We argue that a key factor driving this disconnect is the nature of public involvement; in particular, a near-absence of *meaningful* public involvement through which affected residents can influence and shape decisions. We propose a new model for public involvement in transportation planning, based on a review of the literature and two policy and planning case studies. Our model identifies addressing the priority unmet needs of underserved communities as a fundamental touchstone of equity and argues for dedicating a stream of funding to meet those needs. Widespread application of this model, at the regional level and elsewhere, would be likely to result in improvements to both the general welfare and transportation equity conditions.

Keywords: public participation, civil rights, justice, advocacy

Introduction

In the United States, federal requirements related to civil rights and environmental justice (EJ) mandate executive agencies to protect minority and low-income populations from unfair treatment and ensure them an opportunity to meaningfully participate in decisions that affect their lives. These requirements are particularly relevant for the regional transportation planning process where substantial sums of public funds are routinely spent and potential social, economic, and environmental impacts can be large in magnitude and long lasting (e.g., Golub, Marcantonio, & Sanchez, 2013). In transportation planning, EJ directives, as well as broader calls for *transportation equity*, have been interpreted to encompass both just process and just outcomes (Karner, Golub, Martens, & Robinson, 2018; T. Sanchez, Stolz, & Ma, 2003). The former refers to the meaningful involvement of those historically burdened by transportation infrastructure—people of color and low-income people—while the latter refers to the fair distribution of the benefits and burdens of transportation plans and programs across space and demographic groups.

Metropolitan planning organizations (MPOs) are the entities responsible for transportation planning and programming activities in regions across the US. They typically undertake an equity analysis of their regional transportation plans, nominally to ensure some degree of outcome justice; however, the problems inherent in MPO analyses are widely known and include the omission of race as a key variable, the elision of current equity gaps in favor of distant forecast year conditions, and inconsistent definitions of affected communities (Karner, 2016; Karner & London, 2014; Karner & Niemeier, 2013; D. Rowangould, Karner, & London, 2016). Here we focus on the potential for meaningful involvement to offer a path towards transportation equity.

Across different spheres of decision making, the literature is clear that traditional approaches to public involvement rarely provide meaningful opportunities to affect outcomes and rarely can be said to result in improved decisions (e.g. J. Innes & Booher, 2004). The literature does provide some models of *meaningful* engagement that appear well tailored to improving outcomes, but they require facilitating consensus among disparate actors with vastly differing resources, goals, and agendas. This high standard is very challenging to achieve within the time-constrained planning environment (Bailey, Grossardt, & Ripy, 2015). Others suggest that building the capacity of community members to participate meaningfully can improve the quality of decisions made (e.g. Goldfrank, 2011). Capacity building can be achieved in many cases simply by providing appropriate resources to community-based organizations. There is a need to extract what is best from each of these approaches and develop a practical and effective framework for achieving just outcomes through meaningful and equitable involvement in transportation decision making.

Here, we propose a new model for transportation equity grounded in the literature on public involvement effectiveness and further supported by two case studies drawn from California policy and planning. The model emerges from the fundamental premise that a particular plan or program can only be judged to be equitable to the extent that it is responsive to the *needs* of low-income and minority residents. This focus on needs is enshrined in federal law, which requires MPOs to “seek out and consider the needs of ... low-income and minority households.”¹ It is also responsive to the long history of the disproportionate impacts of transportation planning processes, which often failed to consider these needs (Golub et al., 2013; e.g., Mohl, 2004). Rather than proposing an abstract definition of just outcomes, we take an operational approach

that emphasizes “identifying and addressing”² the priority unmet needs experienced by underserved populations today. This approach does not rely solely upon quantitative equity analysis, but emphasizes the meaningful engagement of and deliberation among affected residents in identifying, prioritizing, and addressing their needs with dedicated funds. It points to a type of public involvement fundamentally different from that typically conducted by MPOs.

If the transportation planning process is to meaningfully address the needs of underserved residents, we argue that a three-step process is required: (1) those needs and priorities must be identified through a community-led and appropriately resourced decision-making process, (2) they must be addressed with a dedicated near-term revenue stream, and (3) progress in meeting them must be tracked using appropriate metrics over time.

To develop this three-step approach we first review the literature on the place of public involvement in transportation equity to motivate the need for a new framework. We then describe two exemplary efforts in California that demonstrate its practicality. In the final section, we propose our new model for transportation equity based on identifying and meeting the needs of underserved communities. We suggest that the widespread adoption of this model, at MPOs and beyond, would go much further than prevailing practice towards closing critical equity gaps—including spatial mismatch, public health disparities, displacement, and ongoing residential segregation—leading to measurable improvements in equity conditions and greater overall welfare across the country.

Public involvement literature and practice

EJ requires the “full and fair participation by all potentially affected communities,” both as a democratic end in itself and as a means to ensure against an unfair distribution of the benefits and burdens in the decision-making process (Federal Transit Administration, 2012, p. 2). However, despite widespread acknowledgement of its importance, the outputs of traditional public involvement efforts often have the potential to be ignored when the outcomes of a plan are decided upon.

In this literature review, we briefly trace the history of public involvement in transportation planning before turning to the public involvement literature, which confirms that current methods for engaging underserved communities do not work well while offering diverse opinions about how to design effective processes. Finally, we summarize the literature calling for new methods of public involvement to replace traditional approaches but find them wanting. This need for new approaches motivates our investigation of two exemplars from practice in the section that follows.

Transportation equity and public involvement: A brief history

The contemporary movement for transportation equity can be traced to the construction of interstate freeways following the 1956 Federal-Aid Highway Act and the contemporaneous civil rights struggles to end Jim Crow segregation in public transit systems that culminated in the Montgomery Bus Boycott (Bullard, 2004). Freeway construction resulted in the widespread displacement of people of color (especially Black) and low-income people (see, e.g., Keating, 2001; Self, 2003) while an affluent white population benefitted from new suburban subdivisions and connections to jobs in central business districts (e.g., Dyble, 2007).

Subsequent freeway construction quickly spawned “revolts” in cities across the US (Dyble, 2007; Mohl, 2004), laying bare the need for meaningful public involvement. In the early

interstate era, participation was limited to a single hearing, held after a preferred alignment had been chosen, meant to inform a community about pending construction. The single hearing was later expanded to two (Weiner, 2008). A “decide, announce, defend” mentality among engineers and transport officials was pervasive at the time (e.g., Irving, 2003). Residents at risk of displacement had good reason to believe their voices were not being heard in this process.

The disparate and adverse impacts of freeway construction on people of color and low-income people demonstrated that transportation resources were not distributed equitably across population groups and that mitigating policy was required. Because of evident racial disparities, new civil rights legislation, like Title VI of the 1964 Civil Rights Act, was applied to transportation. In 1970, the US Department of Transportation issued its first regulations to implement Title VI, and participation was a key category of protected activities (Martens & Golub, 2014). One year earlier, the National Environmental Policy Act (NEPA) required federal agencies to disclose environmental impacts to the public (Dietz & Stern, 2008, p. 37). Subsequent guidance related to EJ came in a 1994 Executive Order, requiring all executive agencies to “identify and address, as appropriate, disproportionately high and adverse human health or environmental effects of its programs, policies, and activities on minority populations.”³ Related EJ guidance incorporates the need to “ensure full and fair participation by all potentially affected communities in the transportation decision-making process” and to “prevent the denial of, reduction in, or significant delay in the receipt of benefits by minority and low-income populations” (Federal Transit Administration, 2012).

Yet despite these policy changes in response to historical excesses, the metropolitan segregation and exclusion of people of color and low-income people in the US continues apace. Disparities abound in the areas of collision-related mortality and injury rates (Morency, Gauvin, Plante, Fournier, & Morency, 2012), roadway proximity (G. M. Rowangould, 2013), public health risks (Rosenbaum, Hartley, & Holder, 2011), access to employment (especially when relying upon public transportation) (Golub & Martens, 2014; Grengs, 2010), and opportunities for quality food, housing, and education (Grengs, 2015; Schill & Wachter, 1995). Part of the explanation for ongoing disparities lies in the contested nature of public involvement’s purpose.

Measuring the effectiveness of public involvement

The contested nature of public involvement in the wake of this history is reflected in a diversity of views on its purpose in the academic and applied literatures within and beyond planning. For example, Innes and Booher (2004) enumerate five purposes: 1) discovering public preferences, 2) improving decisions through the incorporation of local knowledge, 3) advancing fairness and justice by addressing the concerns of traditionally underrepresented groups, 4) enhancing the legitimacy of decision making, and 5) legal compliance (see also Morris & Fragala, 2010). Those authors note that in an effort to guard against liability, agencies often focus on the fifth purpose without adequately considering the other four. Fiorino (1990) argues that members of the public possess valuable knowledge not available to decision makers, their inclusion supports the ideals of democracy, and effective participation increases the legitimacy of the process and improves outcomes. Rowe & Frewer (2000) provide further support for these arguments, noting that experts do not always agree and they have historically ignored or downplayed key pieces of evidence related to particular risk management decisions.

Understanding how well these various objectives are met in practice requires the development of performance measures that can be applied to assess a given public involvement

effort. Webler (1995) argues that only normative performance measures are appropriate for use in part because perceptions of the quality of a particular process will vary from person to person. He proposes “fairness and competence” as the normative standards against which all efforts should be assessed. Fairness refers to the absence of power differentials across participants and competence refers to the ability for a particular process to arrive at a shared worldview. Rowe & Frewer (2000) elaborate upon Webler’s framework and helpfully delineate two types of normative performance measures: acceptance criteria (relating to the public acceptability or perceived fairness of a given effort) and process criteria (relating to its potential to be effective and mostly concerning the resources available to participants) (see also, Rowe & Frewer, 2005). Abelson et al. (2003) propose four categories of measures: stakeholder representation, quality of the procedures, the type of information used, and effects on outcomes.

In contrast, and given the inherent difficulties with conducting an assessment detached from the experiences of participants, other authors have proposed explicit performance measures that can be widely applied across different public involvement exercises and which consider the experiences of participants as a key data source. For example, Bailey et al. (2015) propose a framework based on quality, inclusion, clarity, and cost-effectiveness, based in part on detailed field experiments and data collection (Bailey & Grossardt, 2010; Bailey, Grossardt, & Ripy, 2012). Their methods for operationalizing quality depend upon real-time polling of participants at public meetings asking about their satisfaction with the process.

Many of the performance measures identified by prior authors overlap conceptually and address both process- and outcome-related factors. One factor that emerges across many different studies is that of influence—that a “genuine impact on policy” will emerge from a given process (Rowe & Frewer, 2000, p. 14). Rowe & Frewer (2000) suggest that influence can be achieved by stating at the outset how public involvement will be used to affect policy. This true outcome measure is consistent with the broader goals for public involvement efforts discussed earlier.

Rowe & Frewer (2004) make an explicit distinction between process and outcome performance measures, noting that it is often necessary to use process-related measures to assess performance because of the difficulty of measuring the latter (p. 520). Other authors concur, noting that typical performance measures include counts of the number of meetings and their attendees, determining whether participants are representative of the larger community, and documenting forms of communication provided (Beierle, 1998; Morris & Fragala, 2010).

Assessing the impact of public involvement on outcomes is challenging because outcomes may manifest long after a particular involvement effort is completed, and disentangling cause and effect in complex policy and planning process with many inputs, outputs, and near-term decision points is not straightforward (Bickerstaff & Walker, 2005). But some authors have shown a link between meaningful participation and outcomes in the area of environmental health through in-depth study. Freudenberg et al. (2011) suggest that building community capacity through measures like meaningful public involvement, data collection, and analysis can reduce local environmental burdens faced by communities. Corburn (2005) reports similar results.

Despite the potential difficulties of measuring influence, the importance of connecting a particular process to just outcomes has been present in the public involvement evaluation literature from the beginning. Writing during the height of urban renewal and interstate highway construction, Arnstein (1969) proposed a typology of public involvement efforts ranging from

those processes with the mere appearance of involvement (e.g., “manipulation”) to those entailing real community control. Her implicit indicator of performance was the extent to which members of the public controlled—or were able to affect the outcomes of—a process.

Allowing that appropriate expert input is likely to be desirable in most circumstances (e.g., Bailey & Grossardt, 2006), the concept of “meaningful involvement” has emerged to describe the potential for participants to shape decisions and achieve just outcomes.⁴ As described by Aimen and Morris (2012, p. 5-1), meaningful involvement requires

that potentially affected community stakeholders and residents have an opportunity to participate in decisions about a proposed activity that will affect their environment, safety, or health. It means that affected communities have an opening to influence government decisions and that all involved participants will be considered in the decision-making process.

A focus on outcomes is important for several reasons. A given public involvement effort can appear to comport with key process-related performance measures like inclusion or early involvement but differentials in decision-making and agenda-setting power can easily lead to perpetuation of the status quo (Connelly, 2006; King, Feltey, & Susel, 1998). Connelly (2006) argues that to achieve effective public involvement, interested parties must become engaged in shaping the very process by which a given effort is designed and implemented, otherwise its outcomes will simply reflect the desires of the public officials who created it.

In a transportation equity or EJ context, ensuring fair outcomes is an absolute necessity and likely requires proactively identifying and redressing existing inequities (Golub, Marcantonio, & Sanchez, 2013; Karner & Niemeier, 2013; Martens, Golub, & Robinson, 2012). In the transportation planning sphere, where substantial sums of money are at stake and a plan comprises hundreds of multimodal projects, best practices for public involvement often entail a number of public meetings in which participants are asked to choose between alternative transportation and land use scenarios pre-defined by the agency (e.g., Niemeier, Grattet, & Beamish, 2015). But when judged against the high standards set by the EJ movement, these efforts clearly fall short. According to Cole & Foster (2001), an effective public involvement process should head off the worst of future disparate impacts arising from transportation planning activities, by bringing issues and preferences to light early on in the planning process, surfacing alternatives, and allowing affected communities a chance to speak for themselves. London et al. (2008) and Shilling et al. (2009) report on broader EJ policy implementation in California, where EJ policies and practices are widely institutionalized and advocacy organizations relatively strong compared to other locations in the US. They conclude that state agencies emphasize process-related factors like inclusion but provide no guarantee of substantive effects on outcomes.

To be clear, achieving the ideals of meaningful involvement and outcome justice are not likely to be possible without a just process. If there are too few participants or they are not broadly representative of the groups whose participation is most needed, the process will not be successful. Ensuring process justice is difficult. Public participation processes often fall short of success because the benefits of participating (potential to impact decisions) do not outweigh the costs (e.g., self-study, time, travel, public speaking). This point—that an increase in the benefits and an explicit link between participation and outcomes as strategies to improve participation—has been made across a spectrum of researchers, from public choice theory (Rydin & Pennington, 2000) to participatory budgeting (PB). PB – which emerged in Brazil in the late 1980s before spreading to other locations in Latin America, Europe and beyond (Cabannes,

2004)—“allows the participation of non-elected citizens in the conception and/or allocation of public finances” (Sintomer, Herzberg, & Röcke, 2008, p. 168).⁵ Sintomer et al. (2008) emphasize that, in addition to process-related elements (repeated public meetings specific to the allocation of resources), outcomes must also be tracked so that the overall PB effort embodies accountability. Comparative assessments in the PB literature demonstrate that community control results in dramatically greater and more meaningful engagement than a merely advisory voice (Goldfrank, 2011). Regional transportation planning and programming offers fertile ground for the deployment of PB and related concepts.

Despite the potential benefits of and need for meaningful involvement and outcome justice, there is widespread evidence that current efforts fail to meet this standard. Innes and Booher (2004, p. 419) forcefully make the point that traditional public involvement does not work well:

Legally required methods of public participation in government decision making in the US...do not work. They do not achieve genuine participation in planning or other decisions; they do not satisfy members of the public that they are being heard; they seldom can be said to improve the decisions that agencies and public officials make; and they do not incorporate a broad spectrum of the public.

The need for new models and methods of engagement

Recognizing the failure of traditional efforts to rise to the “meaningful” standard, a number of authors have called for entirely new methods of public involvement whose performance is not evaluated with reference to process-related measures (Bickerstaff & Walker, 2001). Innes and Booher (2004) propose a collaborative model of participation as an antidote. Their model “is not one-way communication from citizens to government or government to citizens. It is a multi-dimensional model where communication, learning and action are joined together and where the polity, interests and citizenry co-evolve” (p. 422). Collaborative approaches encompass genuine dialogue across a group of participants representative of all potential stakeholders, early involvement prior to key decision points, and the co-generation of knowledge.

While the promise of these collaborative models is high, and they could be very effective in promoting longer-term regional consensus building (e.g., Frug, 2001), their successful execution requires a number of interrelated pieces to fall into place. In addition to the involvement of professional facilitators, Innes (2004) outlines eight criteria⁶ that must be met to achieve successful outcomes from consensus building processes; and consensus building is simply one component of collaborative planning. Further, to account for power and resource differentials across groups, collaborative planning proponents argue for the provision of technical assistance and additional resources to facilitate the participation of historically disadvantaged groups.

While we do not doubt the potential for the collaborative model to generate positive outcomes in certain situations, especially over the longer term, its execution in near-term decision-making is likely to be challenging. The extended time periods involved in both building the capacity to undertake collaborative processes and establish the prerequisites for involvement also preclude their application to the transportation planning process, given the diversity of stakeholder interests, preferences, and backgrounds (Bailey, Grossardt, & Ripy, 2015).

In sum, meaningful public involvement is a necessary condition for achieving transportation equity, but standard approaches for engaging the public are simply not working well. The simpler model we propose emphasizes achieving meaningful improvements in the conditions faced by disadvantaged populations by addressing process and outcome justice simultaneously. It

is inspired by the public involvement literature cited above and experiences with transportation planning and innovative California legislation.

In the next section, we examine two emerging exemplary policy and planning efforts that embody principles suggested by PB efforts. The success of these efforts suggests that their key principles may be worthy of replication within the transportation planning sphere. In the subsequent section, we synthesize key premises to propose an explicit new framework for transportation equity grounded in meaningful public involvement.

Exemplars from practice

Additional data for the development of a new framework comes from two case studies of practice, both inspired by California climate change policy under California Assembly Bill (AB) 32. Enacted in 2006, the bill authorized the California Air Resources Board (CARB), to implement policies aimed at reducing greenhouse gas emissions (GHGs) in the state to 1990 levels by 2020. In response, CARB implemented a cap-and-trade mechanism (London et al., 2013). In recognition of the substantial share of GHG emissions attributable to transportation, a 2008 companion bill, Senate Bill (SB) 375, requires MPOs to achieve a portion of that GHG reduction by coordinating regional planning for transportation, land use, and housing (Barbour & Deakin, 2012).

The first case is drawn from the distribution of cap-and-trade auction revenues, which exceeded \$2 billion in FY2016. A state law known as SB 535 requires 25% of those revenues to be spent each year on GHG-reducing investments that benefit “disadvantaged communities” (Truong, 2014). CARB’s final implementing guidance specifies the kind of “benefits” that must be provided as those that, among other things, meet the needs of underserved residents.

The second case is inspired by SB 375. By promoting transit-oriented development as a climate-protection strategy, the law holds out the promise of addressing longstanding problems with regional equity by focusing transportation investments in the densest central city areas. That case illustrates that when underserved communities are able to deliberate together about their priority needs, the results not only benefit them but they are superior for the entire region. From both cases, we extract a number of principles that are broadly applicable to transportation investment planning and public participation and apply them to develop an explicit, three-step model for progressing towards transportation equity.

California cap-and-trade auction revenue allocations

In adopting an annual 25% set-aside of cap-and-trade auction revenues to benefit disadvantaged communities, the California Legislature recognized both “the disproportionate impacts climate change will have on disadvantaged and low-income communities in California” and “the potential vulnerability of California’s low-income and disadvantaged population to efforts to reduce greenhouse gas emissions.”⁷ Implementing SB 535’s intent, however, required a more precise determination of which investments would “benefit” disadvantaged communities (Tu & Marcantonio, 2017).

As construction of the early interstates demonstrated, the simple geographic proximity of an investment to a disadvantaged community does not by itself demonstrate a benefit (Mohl, 2004; Rose & Mohl, 2012). A contemporary example is the Bay Area Rapid Transit (BART) District’s \$500 million Oakland Airport Connector. Citing BART’s failure to assess the project’s civil rights and equity implications,⁸ the Federal Transit Administration withdrew \$70 million in

federal funding (Marcantonio & Mayer, 2010). Despite traversing a heavily minority and low-income corridor, the project provides no intermediate stations and primarily benefits those traveling between the Oakland International Airport and the regional rail system, not the people of color living along the alignment.

Instead of relying on the mere proximity of an investment to a disadvantaged community, EJ and civil rights advocates in California identified four questions that would have to be answered in the affirmative for a project to be eligible for SB 535 funding:⁹

1. Does the investment meet an important need identified by a disadvantaged community?
2. Is the benefit a significant one, rather than incidental to other primary benefits?
3. Are the benefits targeted to low-income residents?
4. Does the investment avoid substantial harms, like displacement or exposure to toxics?

CARB incorporated three of these into its final guidance on the allocation of cap-and-trade revenues in December 2015.¹⁰ For instance, it requires agencies allocating cap-and-trade funds meant to be counted against the SB 535 set aside to provide “a benefit that meaningfully addresses an important community need,” and requires that the benefits be “direct, meaningful, and assured” (CARB, 2015, p. 2-6). CARB notes several methods that can be used to identify community needs, including traditional community meetings. It also provides an example table of “commonly identified” community needs presented to it by community groups during CARB’s public involvement process surrounding the development of the guidelines encompassing three categories of needs: public health and safety, economic, and environmental health (Table 1). These were defined ahead of time, without robust public involvement, simply because of the short timelines associated with the development of the guidelines. Table 1 vividly illustrates that the priority needs of underserved communities deviate markedly from those typically pursued by public agencies (e.g., cost-effectiveness, positive benefits in a cost-benefit framework, or Pareto optimality), providing further justification for their robust engagement.

[Table 1 about here.]

The final guidance also contains strong language related to benefits and burdens, consistent with federal EJ policy, requiring agencies to prioritize investments that maximize benefits and to favor those investments with the “most significant” benefits (CARB, 2015, p. 2.A-6). In terms of harms, it requires that “projects be designed to avoid substantial burdens, such as physical or economic displacement...or increased exposure to toxics or other health risks” (CARB, 2015, p. 2.12). These principles have successfully guided billions of dollars in investments that meaningfully meet the needs of underserved communities. Importantly, information generated during this process could later be used in the preparation of a NEPA-required environmental impact statement to demonstrate a project’s purpose and need (e.g., Gribbin & Kaleta, 2005).

Advocates have noted that public engagement during the development of the guidelines was well-executed and responsive to their needs, but have raised concerns regarding CARB’s approaches to individual project prioritization and selection (Russak, 2015). At a convening of nearly 100 community-based organizations in late 2015 to discuss SB 535 implementation, advocates highlighted issues with transparency and potential digital divides, since CARB was largely sharing information online (Coalition for Clean Air, 2015). While these and similar organizations have suggested process-related improvements that might be implemented by CARB related to transparency in project selection and targeted technical assistance and funding

to facilitate meaningful engagement (e.g., Callahan & DeShazo, 2014), no comprehensive evaluation of public involvement in SB 535 has been made to date. But the level of interest in the law and its implementation is likely due in part to the ability to affect revenue allocation. One foundation involved in SB 535 advocacy stated, “[SB 535] represents an exceptional opportunity for environmental justice (EJ) issues to be addressed if these monies are invested [to serve] the interests of vulnerable, low-income households” (Russak, 2015, p. 1).

Participatory planning in the San Francisco Bay Area

The second California exemplar is drawn from the adoption of the Bay Area’s 2013 Regional Transportation Plan/Sustainable Communities Strategy, *Plan Bay Area*, the region’s first RTP to incorporate SB 375 requirements to reduce GHG emissions (MTC & ABAG, 2013). Key policies and practices supported by SB 375 include transit-oriented development, increased transit investment, and the creation of pedestrian- and cyclist-friendly environments.

The region’s MPO, the Metropolitan Transportation Commission (MTC), proposed to meet GHG reduction targets by concentrating 80% of projected housing growth in locally nominated Priority Development Areas (PDAs) well served by transit. In part because few affluent suburban jurisdictions volunteered significant areas for growth, most PDAs overlapped heavily with the same underserved locations that MTC identifies as “communities of concern,” the region’s name for communities containing high proportions of low-income people and people of color.

This concentration of projected growth in underserved communities has two important implications. First, crucial opportunities to reduce GHGs by providing workforce housing well-matched to the earnings of local workers in suburban job centers were not exploited even though substantial mismatches exist between concentrations of low-wage jobs and locations of affordable housing across the region (Benner & Karner, 2016).

The second implication relates to the impact of involuntary displacement on communities of color in the Bay Area. Figure 1 illustrates the clear pattern of Black exodus from the largest jurisdictions and historical Black population centers in the Bay Area between 2000 and 2010. The locations that gained population over this time are generally much more peripheral, and include locations far outside of the Bay Area, reflecting a dispersal of the Black population, consistent with the observed suburbanization of poverty nationwide (e.g., Kneebone & Berube, 2013). Residential shifts in the Black population are especially relevant in the transportation context both because of the targeting of Black neighborhoods for urban renewal and the strong connection between minority and poverty status. As poorer populations more likely to use transit are displaced from areas where it is most effective, they are less likely to use it to meet their daily needs. These demographics shifts may ultimately undermine the effectiveness of GHG mitigations.

[Figure 1 about here.]

To provide a counterpoint to the MPO’s preferred plans, a robust advocacy network emerged calling itself the “6 Wins for Social Equity.” The 6 Wins Network is a regional coalition of over 20 organizing and advocacy groups focused on social, economic and environmental justice. It came together to engage across multiple issue areas in the regional planning process, seeking high quality transit service, affordable housing, healthy and safe communities, protection against displacement, and quality job creation. Specifically, it sought to bring investment back to

underserved communities in order to provide lasting climate mitigation *and* social equity while simultaneously unlocking housing in historically resistant suburban jurisdictions.

Through a community-driven process completely outside of the MPO's official channels, the 6 Wins developed an alternative transportation and land use scenario entitled "Equity, Environment and Jobs" (EEJ). The EEJ scenario differed from the agency's preferred plan by shifting some funding from transit capital to operating and some housing growth from PDAs to suburban job centers. It also conditioned receipt of funding allocated under the "One Bay Area" grant program on local governments' adoption of anti-displacement protections.

MTC ultimately analyzed the EEJ scenario as part of the required environmental impact review process. The performance assessment demonstrated that the alternative generated by low-income communities of color specifically to meet their priority needs outperformed the alternatives and resulted in reduced displacement risk, greater rent savings, and fewer air pollution emissions (Marcantonio & Karner, 2014). The agency also designated the EEJ the "environmentally-superior alternative" among the five scenarios it analyzed in the state-required environmental impact report, while finding that its superior performance included a host of non-equity related performance metrics. While the agency did not adopt the community's plan, it adopted several amendments in recognition of its superior performance.

A new model for transportation equity

Taken together, the two cases highlight several important principles that can be brought to bear on a new model for achieving transportation equity by re-envisioning public involvement. In both cases, placing meaningful public involvement and the self-identification of community needs at the center of the process was vital to achieving success. Under SB 535, fiscal resources are allocated to affirmatively meet those needs in the near term. Both SB 535 and the EEJ case demonstrate that involving community-based organizations in the design of key elements of the underlying planning efforts—project selection criteria in SB 535 or the broad contours of transportation and land use plans in the EEJ—can support just outcomes.

Drawing from these cases and the literature review, we propose a new model for transportation equity that can in principle be applied at the plan, program, or project level. The model, in which a portion of funding is set aside to meet the needs prioritized by community residents, is composed of three steps:

Step 1: Identify current priority unmet needs in disadvantaged communities, providing resources to facilitate meaningful engagement through a PB-like process

Achieving transportation equity requires meeting the needs of underserved communities. Both SB 535 and the EEJ scenario emphasized identifying and addressing these needs. SB 535 guidance provides examples of needs and the EEJ scenario embodied such needs through its proposed transportation investments and land use vision. Despite the emphasis in regional transportation planning on the long-range transportation plan, Step 1 requires a focus on *current* transportation needs and equity gaps, consistent with insights from PB. A new long-range regional transportation plan is adopted every four or five years, depending on air quality attainment status. The most important question about the plan's equity is not how it is projected to perform in the distant horizon year, but rather how well it will meet identified needs in the near term.¹¹ The short-term transportation improvement program (TIP), adopted by MPOs at least every four years, and which ties specific funding sources to projects, can also be analyzed

in this manner; however, that analysis is no substitute for a near-term analysis of the long-range plan, since the TIP includes only a subset of relevant programs and projects and since it is constrained by the long-range plan.

Consistent with the history of the EJ movement and research on transportation equity—and with federal regulations that require MPOs to “seek out and consider the needs of ... low-income and minority households,”¹² our model places meaningful community engagement at the heart of the process. In Step 1, these needs are identified through community deliberation.

Traditional public involvement approaches (e.g. public meetings, focus groups, social media outreach, advisory boards and commissions) may have a place here. But the identification of needs and priorities requires, above all, the opportunity for authentic community deliberation as the 6 Wins case study and the broader public involvement literature illustrates. To achieve that authenticity, agencies should rely on and provide resources to known community-based organizations that have trusting relationships with low-income, transit dependent, senior, and limited English proficiency constituencies. US DOT encourages this practice.¹³

Step 2: Allocate funding to meet the needs identified in Step 1

Setting aside a fair share of funding to address the needs identified in Step 1 dramatically increases the salience and relevance of public involvement for underserved communities, consistent with the findings from PB practice. SB 535 and PB demonstrate the impact of dedicated resources on resident engagement.

MPOs vary widely in their level of discretionary funding (Goldman, 2007), but a range of revenue sources is possible. For instance, nonattainment areas receive funding from the federal government’s congestion mitigation and air quality (CMAQ) program that could be allocated to meet Step 1’s needs. Legislation or a state DOT decision to create a set-aside at the state level is another possibility. Local transportation sales tax measures present another opportunity to set aside funds to meet identified needs of underserved residents. In short, this model can be adapted to virtually any source of public investment.

Step 3: Tailor metrics and measure progress

Once a fair share of funds has been allocated to meet needs the community identifies as its priorities, determining the success of public involvement in providing real benefits to that community becomes much simpler. Metrics can be tailored to quantify the specific changes the community desires, and progress can be measured year by year, as the set-aside funds are expended.

Conducting thoughtful quantitative or qualitative analyses to support these types of benefit and burden determinations is well within the analytical capacity of planning agencies of various sizes. The success of the EEJ Alternative was supported by quantitative travel demand and land use modeling conducted by MTC, but that modeling was not focused on current inequities, and its metrics were not derived from community-identified priorities. SB 535 requires project sponsors to determine whether benefits are meaningful and targeted to disadvantaged communities while avoiding burdens (CARB, 2015). But pinpointing the precise benefits and burdens to be assessed in particular communities will have been part of the public involvement recommended in Step 1. Once appropriately tailored metrics are selected, assessing existing conditions and then tracking near-term change will ensure that the overall model embodies accountability and that its goals are being met.

Conclusions

Dominant conceptions of transportation equity have been based in practice on public involvement strategies of questionable utility and on broadly uninformative quantitative equity analyses. These approaches do little to address the ongoing unmet needs of disadvantaged communities that are at the heart of persistent metropolitan segregation and exclusion. The new three-step model proposed here can be broadly applied at public transportation agencies, including MPOs, public transit agencies and state departments of transportation, and at any other public entity that allocates public dollars.

By learning from the literature on public involvement and real-world case studies drawn from climate change policy and integrated transportation and land use planning, the model avoids many of the challenges inherent in past efforts and embodies the best available knowledge regarding what works by creating a dedicated revenue stream to meet community needs. It is also nimble and adaptable to 3C nature of transportation planning. One potential limitation of the model involves its deviation from standard practice. Indeed, political will and support from community-based organizations will be required to fully implement it across multiple scales of transportation decision making.

The cases identified in this study were drawn from California, a state with a unique history of EJ activism and advocacy and strong community-based organizations. EJ stakeholders were deeply involved in the early efforts to pass Assembly Bill 32 (AB 32), the 2006 law that requires the state to reduce greenhouse gas emissions to 1990 levels by 2020 (London et al., 2013; Sze et al., 2009). This involvement ensured that some principles of EJ, including factors related to procedural justice were included in AB 32 implementation from the beginning. This history undoubtedly led to the emergence of SB 535 and its mandate to direct funds to disadvantaged communities. In the Bay Area, advocates had been seeking more just processes and outcomes for over a decade or more and had brought federal lawsuits and administrative complaints to seek outcome justice (Marcantonio, Golub, Karner, & Dyble, 2017). Following these two cases, multiple state agencies have adopted funding guidelines that explicitly refer to PB as one method by which funds can be allocated to meet the needs of disadvantaged communities (California Department of Transportation, 2017; California Strategic Growth Council, 2017).

An additional vitally important factor that makes the cases possible is funding from nonprofit foundations and the federal government to support the staffing needs and organizing efforts of community-based organizations. The California Endowment, Surdna Foundation, and the Hewlett Foundation have all made grants over the past 10 years targeted explicitly at achieving equitable health outcomes from the integrated transportation and land use planning efforts in the state. The tri-agency Partnership for Sustainable Communities effort (involving EPA, HUD, and DOT) coordinated to disburse approximately \$4.6 billion from 2009-2013 (US Department of Housing and Urban Development, US Department of Transportation, & US Environmental Protection Agency, 2014). The MTC received a \$5 million award used to create a “Regional Prosperity Plan” aimed in part at creating opportunities for low and middle wage earners to live and work near public transit (Metropolitan Transportation Commission, 2014). Similar funding could be used to defray the potentially higher costs associated with the types of public involvement envisioned here relative to more traditional efforts.

Despite California’s unique history and political landscape, there is evidence that the key performance principles outlined here are being adopted outside of California as well. A prominent example is the 2015 rule on “affirmatively furthering fair housing” promulgated by

the US Department of Housing and Urban Development.¹⁴ Its provisions are just beginning to be implemented by HUD grantees, and its effectiveness in practice remains to be assessed. Like the model proposed here, the rule is aimed at mitigating present-day equity challenges related to segregation and access to opportunity through robust community engagement, prioritization of community needs, actions and investments to meet those needs, and measuring progress in the near term. To address continuing challenges with transportation equity, new approaches like those described here must be continuously deployed and refined.

References

- Abelson, J., Forest, P.-G., Eyles, J., Smith, P., Martin, E., & Gauvin, F.-P. (2003). Deliberations about deliberative methods: issues in the design and evaluation of public participation processes. *Social Science & Medicine*, 57(2), 239–251. [https://doi.org/10.1016/S0277-9536\(02\)00343-X](https://doi.org/10.1016/S0277-9536(02)00343-X)
- Aimen, D., & Morris, A. (2012). *Practical Approaches for Involving Traditionally Underserved Populations in Transportation Decisionmaking* (No. NCHRP Report 710). Washington, DC: National Cooperative Highway Research Program.
- Arnstein, S. R. (1969). A ladder of citizen participation. *Journal of the American Institute of Planners*, 35(4), 216–224.
- Bailey, K., & Grossardt, T. (2010). Toward Structured Public Involvement: Justice, Geography and Collaborative Geospatial/Geovisual Decision Support Systems. *Annals of the Association of American Geographers*, 100(1), 57–86. <https://doi.org/10.1080/00045600903364259>
- Bailey, K., & Grossardt, T. H. (2006). Addressing the Arnstein gap: Improving public confidence in transportation planning and design through structured public involvement (SPI). *Proceedings of the 11th International GeoMultimedia Symposium CORP2006*, 337–341.
- Bailey, K., Grossardt, T., & Ripy, J. (2012). Toward environmental justice in transportation decision making with structured public involvement. *Transportation Research Record: Journal of the Transportation Research Board*, (2320), 102–110.
- Bailey, K., Grossardt, T., & Ripy, J. (2015). High-Performance Public Involvement. *Transportation Research Record: Journal of the Transportation Research Board*, 2499, 45–53. <https://doi.org/10.3141/2499-07>
- Barbour, E., & Deakin, E. A. (2012). Smart Growth Planning for Climate Protection. *Journal of the American Planning Association*, 78(1), 70–86. <https://doi.org/10.1080/01944363.2011.645272>
- Beierle, T. C. (1998). *Public participation in environmental decisions: an evaluation framework using social goals*. Retrieved from <http://www.rff.org/rff/documents/rff-dp-99-06.pdf>
- Benner, C., & Karner, A. (2016). Low-wage jobs-housing fit: Identifying locations of affordable housing shortages. *Urban Geography*, 37(6), 883–903. <https://doi.org/10.1080/02723638.2015.1112565>
- Bickerstaff, K., & Walker, G. (2001). Participatory Local Governance and Transport Planning. *Environment and Planning A*, 33(3), 431–451. <https://doi.org/10.1068/a33173>
- Bickerstaff, K., & Walker, G. (2005). Shared Visions, Unholy Alliances: Power, Governance and Deliberative Processes in Local Transport Planning. *Urban Studies*, 42(12), 2123–2144. <https://doi.org/10.1080/00420980500332098>

- Bullard, R. D. (2004). The Anatomy of Transportation Racism. In R. D. Bullard, G. S. Johnson, & A. O. Torres (Eds.), *Highway Robbery: Transportation Racism & New Routes to Equity*. Cambridge, MA: South End Press.
- Cabannes, Y. (2004). Participatory budgeting: a significant contribution to participatory democracy. *Environment and Urbanization*, 16(1), 27–46.
<https://doi.org/10.1177/095624780401600104>
- California Department of Transportation. (2017). *Sustainable Transportation Planning Grant Program FY 2017-2018 Grant Application Guide*. Sacramento, CA: Division of Transportation Planning.
- California Strategic Growth Council. (2017). *Transformative Climate Communities Program: Proposed Final Guidelines*. Sacramento, CA.
- Callahan, C., & DeShazo, J. R. (2014). *Investment Justice through the Greenhouse Gas Reduction Fund*. Los Angeles, CA: UCLA Luskin Center for Innovation.
- CARB. (2015). *Cap-and-Trade Auction Proceeds: Funding Guidelines for Agencies that Administer California Climate Investments*. Sacramento, CA: California Air Resources Board.
- Coalition for Clean Air. (2015). Leading with Equity: Evaluating and Advancing California's Climate Investments. Retrieved from <http://www.ccair.org/leading-with-equity-evaluating-and-advancing-californias-climate-investments/>
- Cole, L. W., & Foster, S. R. (2001). *From the Ground Up: Environmental Racism and the Rise of the Environmental Justice Movement*. New York: New York University Press.
- Connelly, S. (2006). Looking inside public involvement: how is it made so ineffective and can we change this? *Community Development Journal*, 41(1), 13–24.
<https://doi.org/10.1093/cdj/bsi046>
- Corburn, J. (2005). *Street science: Community knowledge and environmental health justice*. The MIT Press.
- Dietz, T., & Stern, P. C. (Eds.). (2008). *Public Participation in Environmental Assessment and Decision Making*. Washington, DC: The National Academies Press.
- Dyble, L. N. (2007). Revolt Against Sprawl: Transportation and the Origins of the Marin County Growth-Control Regime. *Journal of Urban History*, 34(1), 38–66.
<https://doi.org/10.1177/0096144207308049>
- Federal Highway Administration/Federal Transit Administration. (1999). *Implementing the Title VI Requirements in Metropolitan and Statewide Planning*. Washington, DC: US Department of Transportation.
- Federal Transit Administration. (2012). *Circular FTA C 4703.1: Environmental Justice Policy Guidance for Federal Transit Administration Recipients*. Washington, DC: US Department of Transportation.
- Fiorino, D. J. (1990). Citizen Participation and Environmental Risk: A Survey of Institutional Mechanisms. *Science, Technology, & Human Values*, 15(2), 226–243.
<https://doi.org/10.1177/016224399001500204>
- Freudenberg, N., Pastor, M., & Israel, B. (2011). Strengthening Community Capacity to Participate in Making Decisions to Reduce Disproportionate Environmental Exposures. *American Journal of Public Health*, 101(S1), S123–S130.
<https://doi.org/10.2105/AJPH.2011.300265>
- Frug, G. E. (2001). *City Making: Building Communities without Building Walls*. Princeton, N.J.: Princeton University Press.

- Goldfrank, B. (2011). *Deepening Local Democracy in Latin America: Participation, Decentralization, and the Left*. University Park, PA: Penn State University Press.
- Goldman, T. (2007). Transportation Tax Ballot Initiatives as Regional Planning Process. *Transportation Research Record: Journal of the Transportation Research Board*, 1997, 9–16.
- Golub, A., Marcantonio, R. A., & Sanchez, T. W. (2013). Race, Space, and Struggles for Mobility: Transportation Impacts on African Americans in Oakland and the East Bay. *Urban Geography*, 34(5), 699–728. <https://doi.org/10.1080/02723638.2013.778598>
- Golub, A., & Martens, K. (2014). Using principles of justice to assess the modal equity of regional transportation plans. *Journal of Transport Geography*, 41, 10–20. <https://doi.org/10.1016/j.jtrangeo.2014.07.014>
- Grengs, J. (2010). Job accessibility and the modal mismatch in Detroit. *Journal of Transport Geography*, 18(1), 42–54. <https://doi.org/10.1016/j.jtrangeo.2009.01.012>
- Grengs, J. (2015). Nonwork Accessibility as a Social Equity Indicator. *International Journal of Sustainable Transportation*, 9(1), 1–14. <https://doi.org/10.1080/15568318.2012.719582>
- Gribbin, D. J., & Kaleta, J. S. (2005). *Memorandum on Integration of Planning and NEPA Processes*. Washington, DC: Federal Highway Administration.
- Innes, J., & Booher, D. (2004). Reframing public participation: strategies for the 21st century. *Planning Theory & Practice*, 5(4), 419–436.
- Innes, J. E. (2004). Consensus Building: Clarifications for the Critics. *Planning Theory*, 3(1), 5–20. <https://doi.org/10.1177/1473095204042315>
- Irving, L. (2003). A New Approach to Road Building. *Public Roads*, 67(1), 18–21.
- Karner, A. (2016). Planning for transportation equity in small regions: Towards meaningful performance assessment. *Transport Policy*, 52, 46–54.
- Karner, A., Golub, A., Martens, K., & Robinson, G. (2018). Transportation and environmental justice: history and emerging practice. In R. Holifield, J. Chakraborty, & G. Walker (Eds.), *The Routledge Handbook of Environmental Justice*. New York: Routledge.
- Karner, A., & London, J. (2014). Rural Communities and Transportation Equity in California's San Joaquin Valley. *Transportation Research Record: Journal of the Transportation Research Board*, (2452), 90–97.
- Karner, A., & Niemeier, D. (2013). Civil rights guidance and equity analysis methods for regional transportation plans: a critical review of literature and practice. *Journal of Transport Geography*, 33, 126–134. <https://doi.org/10.1016/j.jtrangeo.2013.09.017>
- Keating, L. (2001). *Atlanta: Race, Class, and Urban Expansion*. Philadelphia, PA: Temple University Press.
- King, C. S., Feltey, K. M., & Susel, B. O. (1998). The Question of Participation: Toward Authentic Public Participation in Public Administration. *Public Administration Review*, 58(4), 317–326.
- Kneebone, E., & Berube, A. (2013). *Confronting Suburban Poverty in America*. Washington, D.C: Brookings Institution Press.
- London, J., Karner, A., Sze, J., Rowan, D., Gambirazzio, G., & Niemeier, D. (2013). Racing climate change: Collaboration and conflict in California's global climate change policy arena. *Global Environmental Change*, 23(4), 791–799. <https://doi.org/10.1016/j.gloenvcha.2013.03.001>

- London, J., Sze, J., & Liévanos, R. S. (2008). Problems, Promise, Progress, and Perils: Critical Reflections on Environmental Justice Policy Implementation in California. *UCLA Journal of Environmental Law and Policy*, 26(2), 255–289.
- Marcantonio, R., Golub, A., Karner, A., & Dyble, L. N. (2017). Confronting Inequality in Metropolitan Regions: Realizing the Promise of Civil Rights and Environmental Justice in Metropolitan Transportation Planning. *Fordham Urban Law Journal*, 44.
- Marcantonio, R., & Karner, A. (2014). Disadvantaged Communities Teach Regional Planners a Lesson in Equitable and Sustainable Development. *Poverty & Race*, 23(1), 5–12.
- Marcantonio, R., & Mayer, G. (2010). Civil rights coalition challenges unfair transit project. *Race, Poverty & The Environment, Spring*, 34–38.
- Martens, K., & Golub, A. (2014). A Fair Distribution of Transportation Benefits: Interpreting Civil Rights Regulations for Transportation Investment Programs. *Working Paper*.
- Martens, K., Golub, A., & Robinson, G. (2012). A justice-theoretic approach to the distribution of transportation benefits: Implications for transportation planning practice in the United States. *Transportation Research Part A: Policy and Practice*, 46(4), 684–695.
<https://doi.org/10.1016/j.tr.2012.01.004>
- Metropolitan Transportation Commission. (2014). *Economic Prosperity Strategy*. Oakland, CA.
- Mohl, R. A. (2004). Stop the Road: Freeway Revolts in American Cities. *Journal of Urban History*, 30(5), 674–706. <https://doi.org/10.1177/0096144204265180>
- Morency, P., Gauvin, L., Plante, C., Fournier, M., & Morency, C. (2012). Neighborhood social inequalities in road traffic injuries: the influence of traffic volume and road design. *American Journal of Public Health*, 102(6), 1112–1119.
<https://doi.org/10.2105/AJPH.2011.300528>
- Morris, A., & Fragala, L. (2010). *Effective Public Involvement Using Limited Resources* (No. NCHRP Report 407). Washington, DC: National Cooperative Highway Research Program. Retrieved from <http://trid.trb.org/view.aspx?id=927763>
- MTC, & ABAG. (2013). *Plan Bay Area: Strategy for a Sustainable Region*. Oakland, CA: Metropolitan Transportation Commission and Association of Bay Area Governments.
- Niemeier, D., Grattet, R., & Beamish, T. (2015). “Blueprinting” and climate change: Regional governance and civic participation in land use and transportation planning. *Environment and Planning C: Government and Policy*, 33(6), 1600–1617.
<https://doi.org/10.1177/0263774x15614181>
- Rose, M. H., & Mohl, R. A. (2012). *Interstate: Highway Politics and Policy Since 1939*. Knoxville, TN: University of Tennessee Press.
- Rosenbaum, A., Hartley, S., & Holder, C. (2011). Analysis of diesel particulate matter health risk disparities in selected US harbor areas. *American Journal of Public Health*, 101 Suppl 1, S217–223. <https://doi.org/10.2105/AJPH.2011.300190>
- Rowangould, D., Karner, A., & London, J. (2016). Identifying environmental justice communities for transportation analysis. *Transportation Research Part A: Policy and Practice*, 88, 151–162.
- Rowangould, G. M. (2013). A census of the US near-roadway population: Public health and environmental justice considerations. *Transportation Research Part D: Transport and Environment*, 25, 59–67. <https://doi.org/10.1016/j.trd.2013.08.003>
- Rowe, G., & Frewer, L. J. (2000). Public Participation Methods: A Framework for Evaluation. *Science, Technology & Human Values*, 25(1), 3–29.
<https://doi.org/10.1177/016224390002500101>

- Rowe, G., & Frewer, L. J. (2004). Evaluating Public-Participation Exercises: A Research Agenda. *Science, Technology & Human Values*, 29(4), 512–556.
<https://doi.org/10.1177/0162243903259197>
- Rowe, G., & Frewer, L. J. (2005). A Typology of Public Engagement Mechanisms. *Science, Technology & Human Values*, 30(2), 251–290.
<https://doi.org/10.1177/0162243904271724>
- Russak, B. (2015). *Advantaging Communities: Co-Benefits and Community Engagement in the Greenhouse Gas Reduction Fund*. Los Angeles, CA: Liberty Hill Foundation.
- Rydin, Y., & Pennington, M. (2000). Public Participation and Local Environmental Planning: The collective action problem and the potential of social capital. *Local Environment*, 5(2), 153–169. <https://doi.org/10.1080/13549830050009328>
- Sanchez, T., Stolz, R., & Ma, J. S. (2003). *Moving to Equity: Addressing Inequitable Effects of Transportation Policies on Minorities*. Cambridge, MA: The Civil Rights Project at Harvard University.
- Sanchez, T. W., & Wolf, J. (2007). Environmental Justice and Transportation Equity: A Review of MPOs. In R. D. Bullard (Ed.), *Growing Smarter: Achieving Livable Communities, Environmental Justice, and Regional Equity*. Cambridge, MA: MIT Press.
- Schill, M. H., & Wachter, S. M. (1995). Housing market constraints and spatial stratification by income and race. *Housing Policy Debate*, 6(1), 141–167.
<https://doi.org/10.1080/10511482.1995.9521184>
- Self, R. O. (2003). *American Babylon: Race and the Struggle for Postwar Oakland*. Princeton, NJ: Princeton University Press.
- Shilling, F. M., London, J. K., & Liévanos, R. S. (2009). Marginalization by collaboration: Environmental justice as a third party in and beyond CALFED. *Environmental Science & Policy*, 12(6), 694–709.
- Sintomer, Y., Herzberg, C., & Röcke, A. (2008). Participatory Budgeting in Europe: Potentials and Challenges. *International Journal of Urban and Regional Research*, 32(1), 164–178.
<https://doi.org/10.1111/j.1468-2427.2008.00777.x>
- Sze, J., Gambirazzio, G., Karner, A., Rowan, D., London, J., & Niemeier, D. (2009). Best in Show? Climate and Environmental Justice Policy in California. *Environmental Justice*, 2(4), 179–184. <https://doi.org/10.1089/env.2009.0028>
- Truong, V. (2014). Addressing Poverty and Pollution: California’s SB 535 Greenhouse Gas Reduction Fund. *Harvard Civil Rights-Civil Liberties Law Review*, 49, 493.
- Tu, C., & Marcantonio, R. (2017). A framework for equitable investment. *Berkeley Planning Journal*, 28(1), 100–117.
- US Department of Housing and Urban Development, US Department of Transportation, & US Environmental Protection Agency. (2014). *Partnership for Sustainable Communities: Five Years of Learning from Communities and Coordinating Federal Investments*.
- US Environmental Protection Agency. (2016). *EJ 2020 Action Agenda: Environmental Justice Strategic Plan*. Washington, DC.
- Webler, T. (1995). “Right” discourse in citizen participation: an evaluative yardstick. In O. Renn, T. Webler, & P. Wiedemann (Eds.), *Fairness and competence in citizen participation* (pp. 35–77). Retrieved from <http://www.springerlink.com/index/N488012632212M08.pdf>
- Weiner, E. (2008). *Urban Transportation Planning in the United States: History, Policy, and Practice*. New York: Springer.

¹ 23 CFR § 450.316 (a) (1) (vii).

² President Clinton's 1994 Executive Order 12898 requires federal agencies to "identify and address" impacts likely to disproportionately affect populations of color and low income. See Executive Order 12898, "Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations" (59 FR 7629, February 16, 1994), § 1-101.

³ Executive Order 12898, § 1-101.

⁴ The notion of meaningful involvement is also enshrined in the US Environmental Protection Agency's definition of EJ as "the fair treatment and meaningful involvement of all people regardless of race, color, national origin or income with respect to the development, implementation, and enforcement of environmental laws, regulations and policies" (US Environmental Protection Agency, 2016, p. 1).

⁵ In an MPO context, this is particularly appropriate given that the members of MPO boards nationally are almost never democratically elected nor are they representative of their regional constituents (T. W. Sanchez & Wolf, 2007).

⁶ The criteria include, *inter alia*, "A process that begins with mutual understanding of interests and avoids positional bargaining" and "A self-organizing process unconstrained by conveners in its time or content and which permits the status quo and all assumptions to be questioned" (J. E. Innes, 2004, p. 7).

⁷ California SB 535 (2012), Sec. 1.

⁸ See letter of FTA Administrator Peter Rogoff to Steve Heminger and Dorothy Dugger, Feb. 12, 2010, available at http://www.publicadvocates.org/wp-content/uploads/feb_12_bart_mtc_letter.pdf.

⁹ A fifth question regarding timeliness was not necessary in this context due to the programmatic requirement that 25% of the revenues benefit disadvantaged communities *each year*.

¹⁰ The one exception was the targeting of benefits to low-income residents. Later legislation, Assembly Bill 1550 (2016), was proposed to require a portion of cap-and-trade revenues to be targeted to "low-income households and communities." As of late 2016 it is awaiting the governor's signature.

¹¹ This horizon year approach, as noted in the introduction, is the focus of much quantitative analysis, but suffers from fundamental flaws that are by now well-documented (Karner & Niemeier, 2013).

¹² 23 CFR § 450.316 (a) (1) (vii).

¹³ In its review of MPO self-certification of Title VI compliance, the joint federal agencies ask "Has the metropolitan planning organization (MPO) or State DOT made funds available to local organizations that represent low-income and minority populations to enable their participation in planning processes?" (Federal Highway Administration/Federal Transit Administration, 1999, pp. 1-2).

¹⁴ Department of Housing and Urban Development, Affirmatively Furthering Fair Housing, 80 Fed. Reg. 42272 (July 16, 2015) (Final rule).

TABLE 1. Commonly Identified Priority Needs of Underserved Communities**Public Health and Safety**

1. Reduce health harms caused by air pollution and suffered disproportionately by low-income residents/communities (e.g., asthma).
2. Reduce health harms related to the built environment suffered disproportionately by low-income residents/communities (e.g., obesity).
3. Increase community safety.
4. Reduce heat-related illnesses and increase thermal comfort.
5. Increase access to parks, greenways, open space, recreation, and other community assets.

Economic

1. Create quality jobs and increase family income (e.g., targeted hiring for living wage jobs that provide access to health insurance and retirement benefits with long-term job retention, community benefit agreements, etc.).
2. Increase job readiness and career opportunities (e.g., workforce development programs, on-the-job training, industry-recognized certifications).
3. Revitalize local economies (e.g., increased use of local businesses) and support California-based small businesses.
4. Reduce housing costs (e.g., affordable housing).
5. Reduce transportation costs (e.g., free or reduced cost transit passes) and improve access to public transportation (e.g., new services in under-served urban and rural communities).
6. Reduce energy costs (e.g., weatherization, solar, etc.).
7. Improve transit service levels and reliability on systems/routes that have high use by low-income riders.
8. Bring jobs and housing closer together (e.g., affordable housing in transit-oriented development and in healthy, high-opportunity neighborhoods).
9. Preserve community stability and maintain housing affordability for low-income households (e.g., prioritize projects in jurisdictions with anti-displacement policies in place).

Environmental

1. Reduce exposure to local environmental contaminants, such as toxic air contaminants, criteria air pollutants, and drinking water contaminants (e.g., provide a buffer between bike/walk paths and corridors with high levels of transportation-related air pollution).
2. Prioritize zero-emission vehicle projects for areas with high diesel air pollution.
3. Reduce exposure to pesticides in communities near agricultural operations.

Source: Adapted from CARB (2015, Table 2-2).

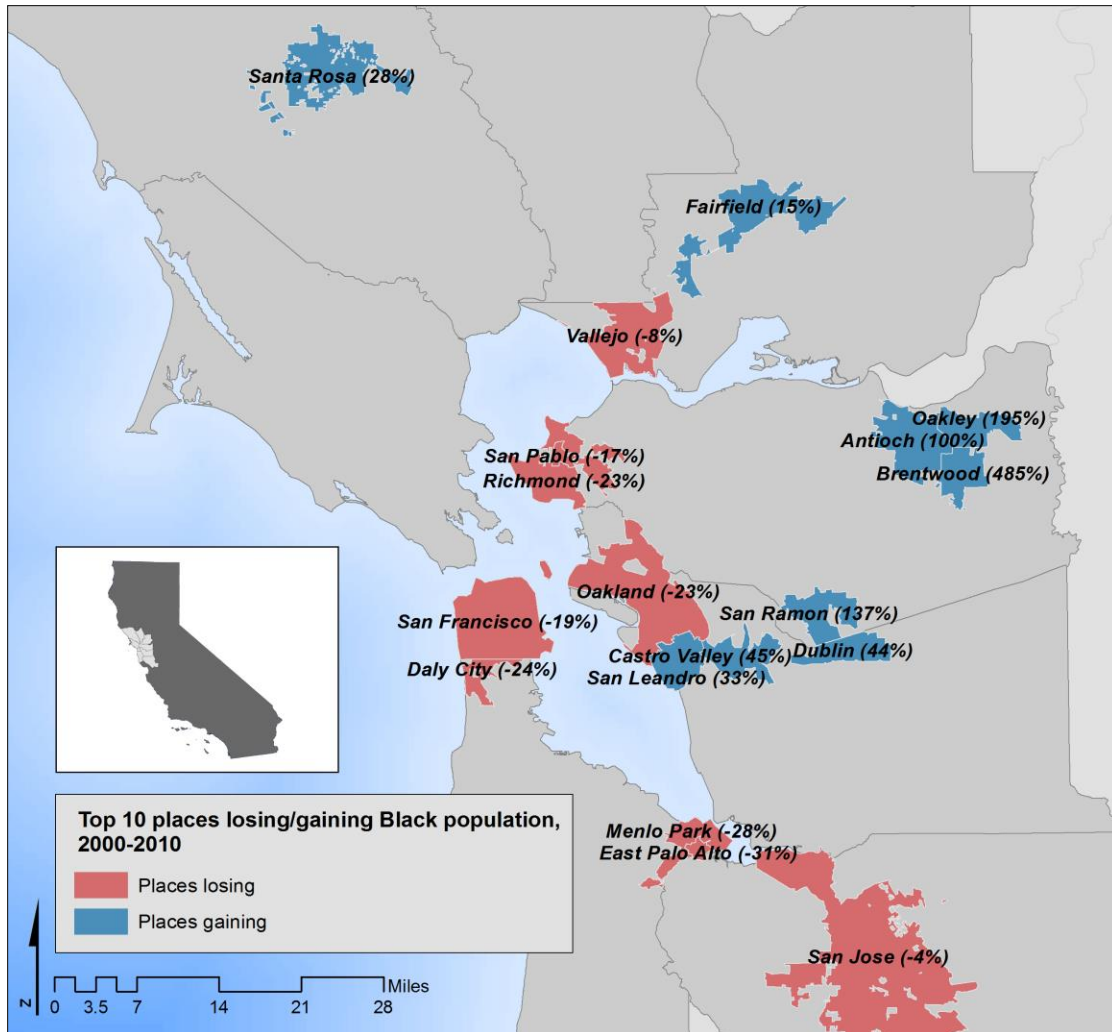


FIGURE 1. Proportion changes in the Black population (of every ethnicity) in Bay Area jurisdictions from 2000-2010. Jurisdictions shown are those 10 experiencing the greatest absolute gains (blue) or losses (red) in Black population (regardless of ethnicity) over this period. Location of the nine-county Bay Area within California shown in inset. Source: US Decennial Census 2000 and 2010.